Ada Lovelace

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Staff software engineer and technical leader with 14 years of experience building reliable developer platforms, distributed systems, and secure delivery infrastructure. Specializes in turning complex operational problems into maintainable products, measurable engineering practices, and clear technical strategy.

Experience

Distinguished Engineer Jan 2024 – Present

Analytical Engines Ltd

- Set technical direction for a platform used by **2,400 engineers** across 18 product groups and reduced median deployment lead time from 46 minutes to 11 minutes
- Designed a multi-region control plane processing 9 million workflow events per day with 99.99% monthly availability
- Led an architecture review program that prevented 14 high-risk production migrations from shipping without rollback plans
- Mentored 12 staff-level engineers and created a promotion framework adopted across a 320-person engineering organization
- Partnered with security and finance teams to reduce annual infrastructure spending by £3.2 million while improving recovery objectives

Principal Software Engineer Mar 2021 – Dec 2023

Analytical Engines Ltd

- Replaced 27 bespoke deployment pipelines with a shared Rust and TypeScript platform, cutting maintenance effort by 6,800 hours annually
- Introduced artifact signing and provenance verification for 100% of production releases across Linux, macOS, and Windows
- Coordinated a zero-downtime database migration involving 4.6 billion records and 38 dependent services
- Reduced incident recurrence by 41% through structured post-incident reviews and automated follow-up tracking
- Published platform standards used by engineering teams in London, Berlin, Hà Nội, and São Paulo

Senior Software Engineer Jul 2017 – Feb 2021

Difference Engine Co

- Built an event-processing service sustaining 120,000 messages per second with a p99 latency below 85 milliseconds
- Improved integration-test reliability from 91.2% to 99.7% by replacing shared environments with isolated ephemeral stacks
- Led 7 engineers through the extraction of a billing domain from a monolith without interrupting monthly invoicing
- Reduced cloud database costs by 38% through workload profiling, tiered retention, and index redesign
- Created an internal course on distributed-systems failure modes completed by 160 engineers over 2 years

Software Engineer Sep 2013 – Jun 2017

Difference Engine Co

- Delivered the company's first self-service continuous-delivery pipeline, increasing release frequency from weekly to 18 times per day
- Migrated 11 services from manually managed virtual machines to reproducible container images
- Added contract tests for 34 external integrations and reduced partner-related production failures by 52%
- Implemented Unicode-safe search supporting names and addresses across 24 languages

Graduate Software Engineer Jul 2011 – Aug 2013

Babbage Systems

- Developed monitoring tools that detected stalled batch processing 23 minutes earlier on average
- Automated release-note generation for 40 internal libraries and saved 6 hours per release cycle
- Fixed date and timezone defects affecting customers in 8 countries

Education

MSc Advanced Computing Sep 2010 – Jun 2011

Imperial College London

- Graduated with distinction and received the departmental award for the highest-scoring distributed systems dissertation
- Dissertation: **Deterministic Replay for Concurrent Event Processing**

BSc Computer Science and Mathematics Sep 2007 – Jun 2010
University of London
- First-class honours; ranked 3rd in a cohort of 186 students
- Coursework included algorithms, compilers, numerical methods, databases, and formal verification

Visiting Student Jan 2009 – Jun 2009
École Polytechnique
- Completed graduate seminars in programming-language semantics and fault-tolerant systems

Projects

Reproducible Release Toolkit Jan 2019 – Present
- Built a cross-platform release pipeline used by 18 open-source projects and more than 4,000 downstream repositories
- Reduced release preparation from 2 hours to 15 minutes with deterministic builds, signed manifests, and automated changelogs
- Maintains compatibility with Rust, Go, JavaScript, and Python packaging ecosystems

Chronicle Event Store May 2020 – Oct 2023
- Designed an append-only event store handling 2.1 terabytes of compressed history with point-in-time reconstruction
- Added property-based tests covering ordering, deduplication, crash recovery, and schema evolution
- Published benchmark methodology and reproducible datasets for 6 storage engines

Accessible Documentation Search Feb 2018 – Dec 2020
- Created a static search index for technical documentation with keyboard navigation and screen-reader announcements
- Improved successful search completion from 72% to 93% in a study involving 48 participants

Temporal Expression Parser 2016 – 2018
- Implemented locale-aware parsing for partial dates, open-ended ranges, and calendar precision
- Covered 1,800 test cases including leap years, time zones, and non-ASCII month names

Publications

Reproducible Builds for Small Engineering Teams Oct 2022
- Published a 42-page practical guide to build isolation, artifact verification, and release automation
- Downloaded more than 28,000 times in its first year

Operating Multi-Region Control Planes Without Losing Sleep Mar 2021
- Presented failure-injection results from 320 controlled experiments across 5 regions
- Received the conference audience award from 1,100 attendees

Schema Evolution in Long-Lived Event Streams Nov 2019
- Co-authored a peer-reviewed paper comparing 7 migration strategies across 12 production datasets
- Released the evaluation harness under an MIT license

A Practical Taxonomy of Deployment Failures 2017
- Analyzed 3,200 failed deployments and identified controls that reduced repeat failures by 44%

Skills

Programming languages
- Rust, TypeScript, Python, Go, SQL, Bash, and Typst

Distributed systems
- Event streaming, consensus fundamentals, idempotency, backpressure, partitioning, replication, and failure injection

Data platforms
- PostgreSQL, SQLite, Kafka, object storage, data retention, online migrations, and performance profiling

Delivery and security
- GitHub Actions, supply-chain security, SLSA provenance, artifact signing, container platforms, and policy as code

Engineering leadership

- Technical strategy, architecture reviews, incident management, mentoring, hiring, roadmaps, and cross-team facilitation

Languages

- English (native), French (professional), Vietnamese (conversational), and Russian (basic)

Board and Advisory Work

Technical Advisor, Open Computing Foundation Jan 2024 – Present

- Advise 9 maintainer teams on release security, contributor workflows, funding priorities, and project governance
- Established a security-response process that reduced median disclosure-to-patch time from 12 days to 3 days

Board Member, London Computing Education Trust May 2020 – Dec 2023

- Oversaw a £1.8 million annual program serving 6,200 students in 73 schools
- Chaired the technology committee and introduced accessibility requirements for all funded learning platforms

Volunteer Mentor, Code Together Sep 2015 – Present

- Mentored 46 early-career engineers through six-month technical and career-development programs
- Designed workshops on debugging, code review, systems thinking, and effective technical writing

Awards and Recognition

Engineering Impact Award, Analytical Engines Ltd Dec 2023 – Dec 2023

- Recognized for organization-wide improvements to deployment safety and developer productivity

Open Infrastructure Community Award Jun 2021 – Jun 2021

- Honoured for sustained maintenance of reproducible release tooling used across multiple ecosystems

Distinguished Paper Award, Systems Practice Conference Nov 2019 – Nov 2019

- Selected from 84 accepted papers for practical impact and reproducibility

Speaking and Teaching

Designing for Failure, Not Just Success 2024 – Present

- Delivered the workshop 8 times to more than 360 engineers with an average participant rating of 4.8 out of 5

Production Readiness for Distributed Services 2021 – 2023

- Created a 6-module internal curriculum covering observability, capacity, recovery, dependency risk, and safe rollouts
- Trained 14 facilitators who delivered the course to 420 engineers

Guest Lecturer, University of London 2018 – 2022

- Taught annual lectures on event-driven architecture, operational design, and engineering ethics

Open Source Contributions

Maintainer, Reproducible Release Toolkit 2019 – Present

- Reviewed 740 pull requests from 126 contributors and published 58 signed releases
- Introduced contributor documentation that reduced first-contribution completion time by 36%

Contributor, Rust ecosystem 2016 – Present

- Contributed fixes and documentation to libraries for serialization, command-line parsing, dates, and structured logging
- Reported and minimized 17 cross-platform compiler and tooling defects

Certifications

Cloud Security Professional 2023 – 2023

- Renewed through continuing education in identity, cryptography, threat modeling, and incident response

Kubernetes Administrator 2020 – 2023

- Completed performance-based certification covering cluster operations, networking, storage, and troubleshooting

Professional Interests

- Dependable developer infrastructure and reproducible software delivery
- Computing history, archival software, and restoration of early programming texts
- Inclusive engineering education and accessible technical documentation
- Privacy-preserving analytics and responsible operational data use